

Task 1: Create User and Group

1. Create group
 - `sudo groupadd musicfans`
2. Create user
 - `sudo useradd streamer`
3. Set password:
 - `sudo passwd streamer`
4. Add user to group
 - `sudo usermod -aG musicfans streamer`
5. Verify
 - `id streamer`

Task 2: Give Current User Sudo Privileges

Update package list:

- `sudo dnf check-update`

Task 3: SSH Key-Based Authentication

Generate SSH Key:

- `ssh-keygen`

Press Enter for every question.

Keys are stored in:

- `~/.ssh/id_rsa`
- `~/.ssh/id_rsa.pub`

View public key:

- `cat ~/.ssh/id_rsa.pub`

Copy key to remote server

If another Linux machine exists:

- `ssh-copy-id username@server-ip`

Login:

- `ssh username@server-ip`

It should not ask for the password.

Disable Password Authentication

Edit SSH configuration

- `sudo vi /etc/ssh/sshd_config`

Find

- PasswordAuthentication yes

Change to

- PasswordAuthentication no

Save.

Restart SSH

- `sudo systemctl restart sshd`

Task 4: SELinux

Check status:

Check detailed status:

Switch to Permissive Mode:

Return to enforcing:

Task 5: Difference between SELinux and AppArmor

SELinux (Security-Enhanced Linux) and AppArmor are Linux security frameworks that provide Mandatory Access Control (MAC) to improve system security. SELinux uses security labels on files, processes, and resources to define detailed security policies, making it more powerful but also more complex to configure. AppArmor secures applications using profile-based rules based on file paths, making it easier to understand and manage. SELinux is the default security module in Red Hat Enterprise Linux (RHEL), while AppArmor is commonly used in Ubuntu and SUSE Linux

Summary :

SELinux and AppArmor both protect Linux systems by restricting what applications can access. SELinux is more secure and flexible because it uses labels, but it is harder to configure. AppArmor is simpler since it works with file paths and is easier for beginners. Both help prevent unauthorized access and improve system security.

Real-World Example :

A food delivery application like Zomato can benefit from SELinux because it protects critical services such as databases, payment processing, and user information. Even if an attacker compromises one application, SELinux limits its access to the rest of the system, helping keep customer data secure.